

STUDY CASE WINE - CLUSTERING

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Step 1: Load Dataset

	Class	Alcohol	Malic_Acid	Ash	Alcalinity_of_Ash	Magnesium	Total_Phenols	Flavanoid
0	1	14.23	1.71	2.43	15.6	127	2.80	3.0
1	1	13.20	1.78	2.14	11.2	100	2.65	2.7
2	1	13.16	2.36	2.67	18.6	101	2.80	3.2
3	1	14.37	1.95	2.50	16.8	113	3.85	3.4
4	1	13.24	2.59	2.87	21.0	118	2.80	2.6
...	...	...	...	...	...	...	...	...
173	3	13.71	5.65	2.45	20.5	95	1.68	0.6
174	3	13.40	3.91	2.48	23.0	102	1.80	0.7
175	3	13.27	4.28	2.26	20.0	120	1.59	0.6
176	3	13.17	2.59	2.37	20.0	120	1.65	0.6
177	3	14.13	4.10	2.74	24.5	96	2.05	0.7

178 rows × 14 columns

Interpretation:

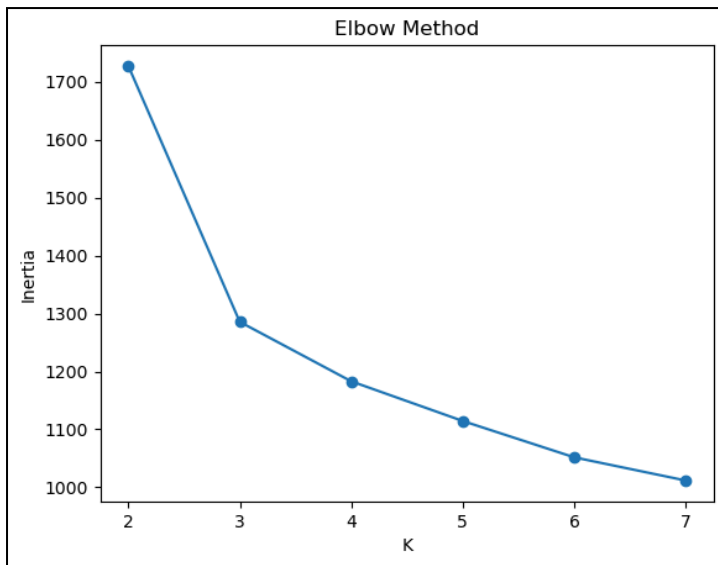
At this stage, the dataset is prepared for unsupervised learning. Since clustering algorithms rely on distance-based calculations, it is essential that all variables are numeric. By selecting only numerical features, the dataset becomes directly compatible with clustering techniques without requiring additional preprocessing such as encoding. This ensures that the structure of the data is preserved and that similarity between observations can be meaningfully computed.

Step 2: Data Scaling (StandardScaler)

Interpretation:

Standardization transforms each feature to have a mean of zero and a standard deviation of one. This step is crucial because clustering algorithms such as K-Means and DBSCAN are sensitive to the scale of the variables. For instance, variables like *Proline* typically have much larger numerical ranges compared to other features. Without scaling, such variables would dominate the distance calculations, leading to biased clustering results. By standardizing the data, all variables contribute equally to the clustering process, allowing the algorithm to identify patterns based on relative differences rather than absolute magnitudes.

Step 3: Determining Optimal Number of Clusters (K)



Optimal K: 3

Interpretation:

Selecting an appropriate number of clusters is a critical step in clustering analysis, particularly for algorithms like K-Means and Agglomerative Clustering that require a predefined number of clusters. The Elbow method evaluates how the within-cluster variance (inertia) decreases as the number of clusters increases, while the Silhouette score measures how well-separated the clusters are. By combining these approaches, the chosen value of K represents a balance between compact clusters and meaningful separation. This prevents underfitting (too few clusters, overly generalized groups) and overfitting (too many clusters, overly fragmented groups).

Step 4: Applying Clustering Methods

Interpretation:

Each clustering algorithm operates under different assumptions and mechanisms. K-Means partitions data into spherical clusters based on centroid distances, while Agglomerative Clustering builds hierarchical relationships between observations. GMM models the data as a mixture of

probabilistic distributions, allowing more flexibility in cluster shapes. In contrast, DBSCAN is a density-based method that identifies clusters based on the concentration of data points and is capable of detecting noise. By applying multiple methods, the analysis ensures a comprehensive comparison, rather than relying on a single algorithm that may not be suitable for the underlying data structure.

Step 5: Clustering Evaluation

Perbandingan Metode	Silhouette	DBI
KMeans	0.307736	1.316396
Agglomerative	0.300591	1.326356
GMM	0.303258	1.330836
DBSCAN	0.358863	1.082149

Interpretation:

The performance of each clustering method is evaluated using two metrics: Silhouette Score and Davies-Bouldin Index (DBI). The Silhouette Score measures how similar an observation is to its own cluster compared to other clusters, with higher values indicating better separation. Meanwhile, the DBI evaluates the average similarity between clusters, where lower values indicate more distinct clusters.

From the results, DBSCAN achieves the highest Silhouette Score (0.358863) and the lowest DBI (1.082149), indicating that it produces clusters that are both well-separated and internally cohesive. In contrast, the other methods show lower separation and higher overlap between clusters. This suggests that the underlying structure of the dataset may not conform well to assumptions such as spherical clusters (as in K-Means), making DBSCAN a more suitable approach.

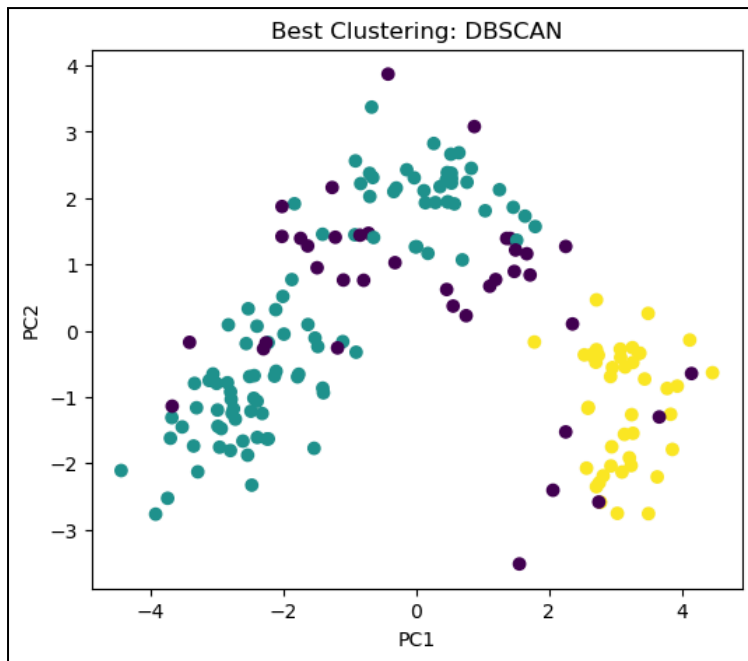
Step 6: Selecting the Best Method

Best Method: DBSCAN

Interpretation:

Based on the evaluation metrics, DBSCAN is selected as the best-performing clustering method. This decision is justified by its superior ability to maximize inter-cluster separation while minimizing intra-cluster variance. Additionally, DBSCAN does not require specifying the number of clusters in advance, making it more flexible in capturing the natural structure of the data. Its density-based approach is particularly advantageous when the dataset contains irregular cluster shapes or noise.

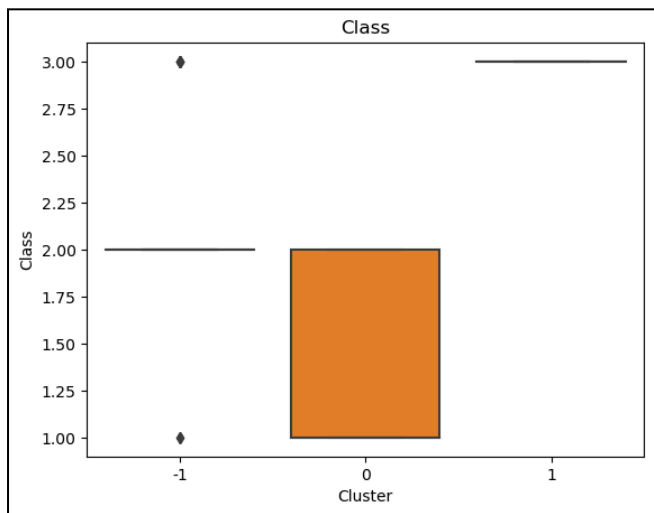
Step 7: Clustering Results

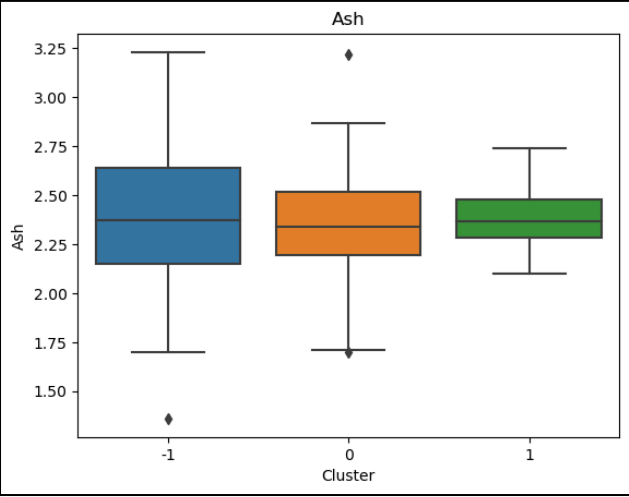
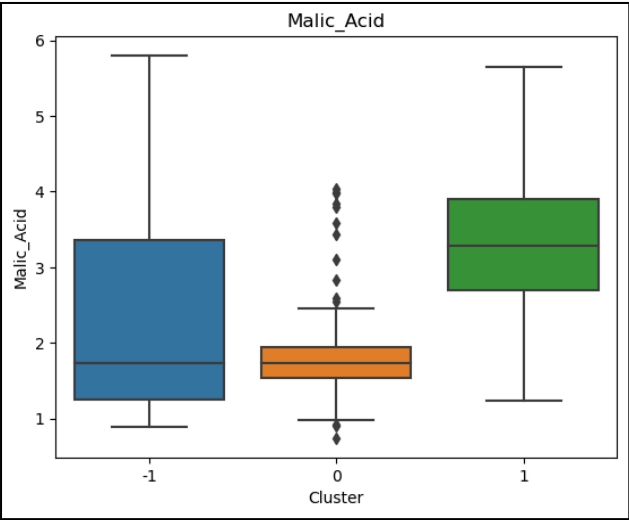
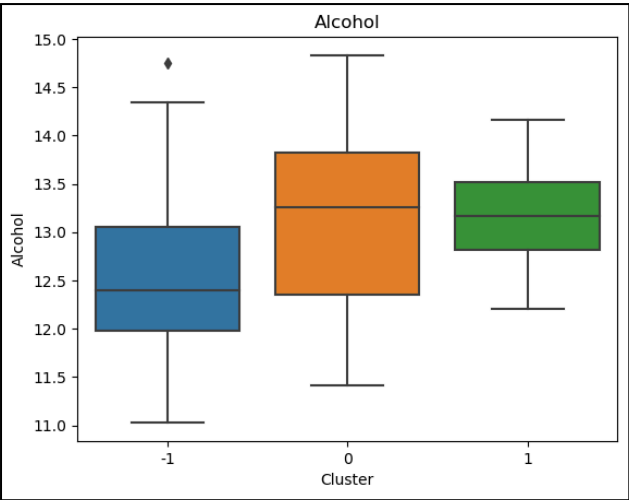


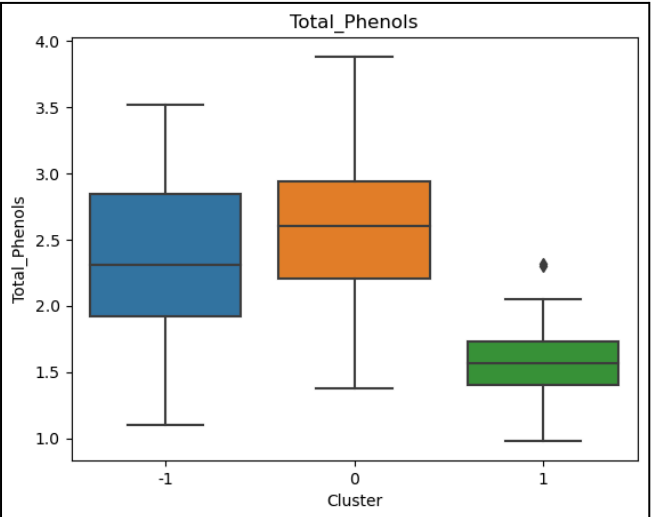
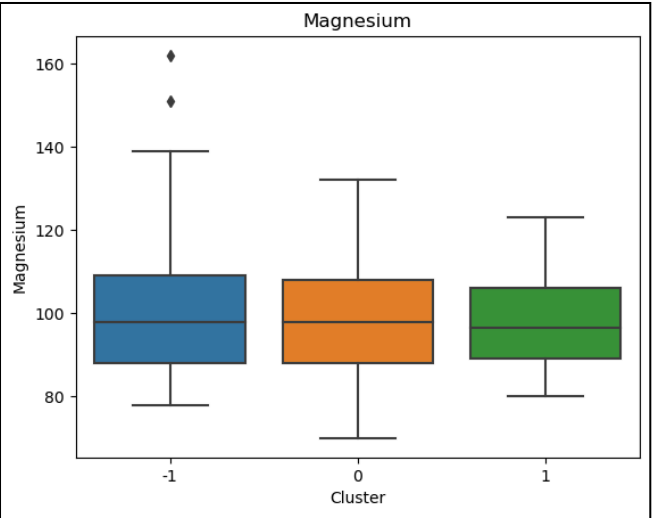
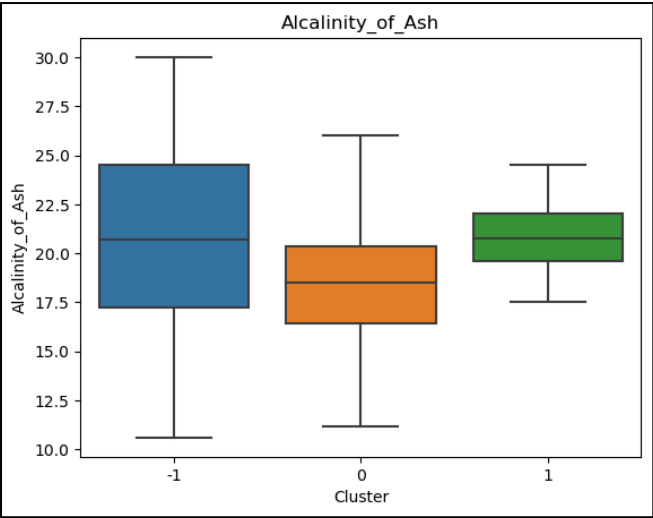
Interpretation:

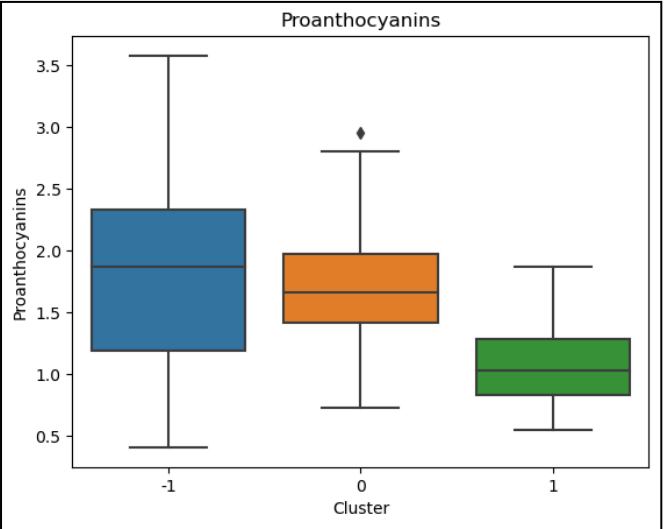
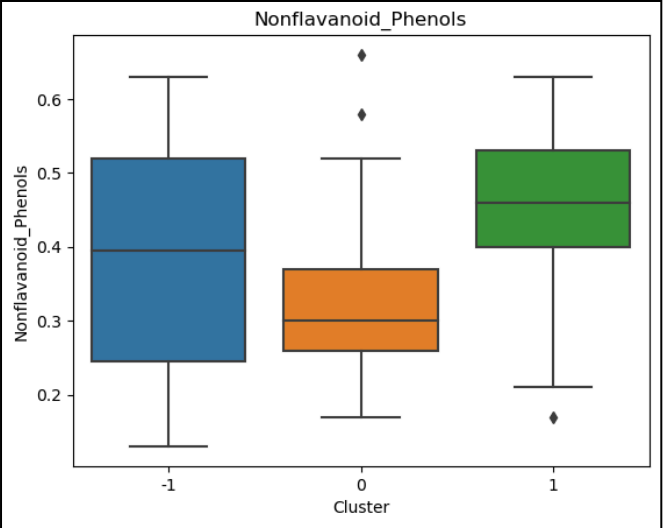
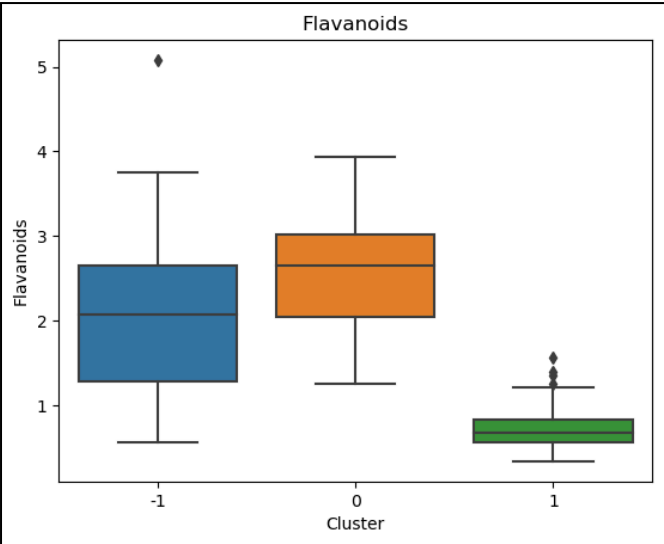
Clusters 0 and 1 represent the main groupings within the dataset, while Cluster -1 represents noise or outliers. Unlike traditional clustering methods, DBSCAN explicitly identifies points that do not belong to any dense region. This is not a limitation, but rather a strength, as it prevents forcing outliers into clusters where they do not naturally belong. The presence of noise indicates that the dataset contains observations with distinct characteristics that differ significantly from the majority.

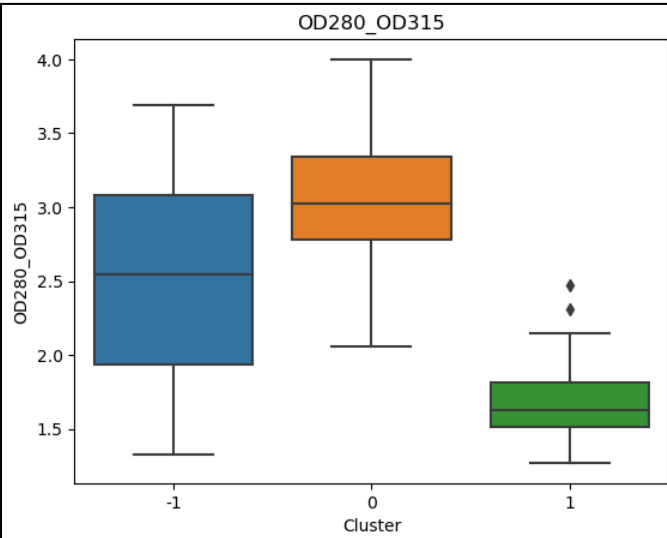
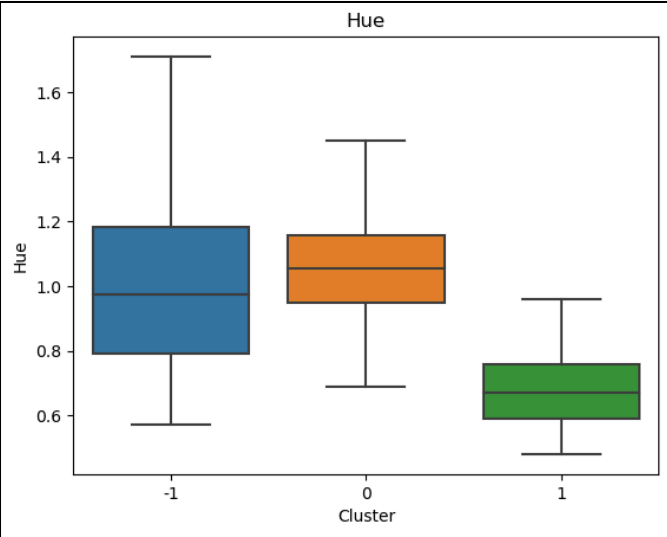
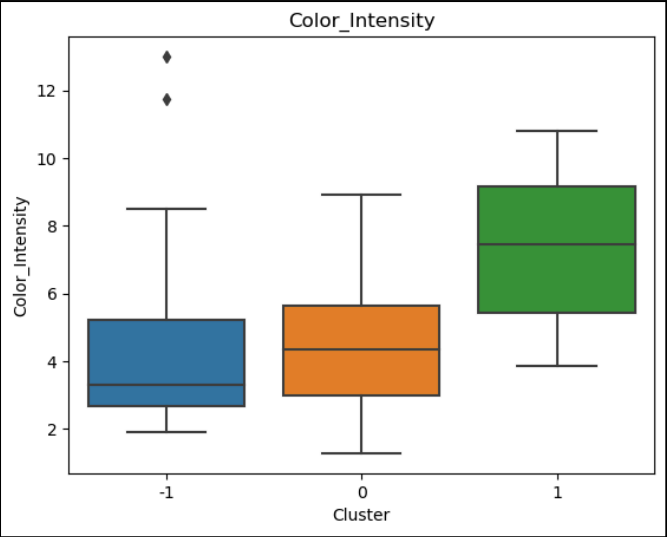
Step 8: Cluster Interpretation

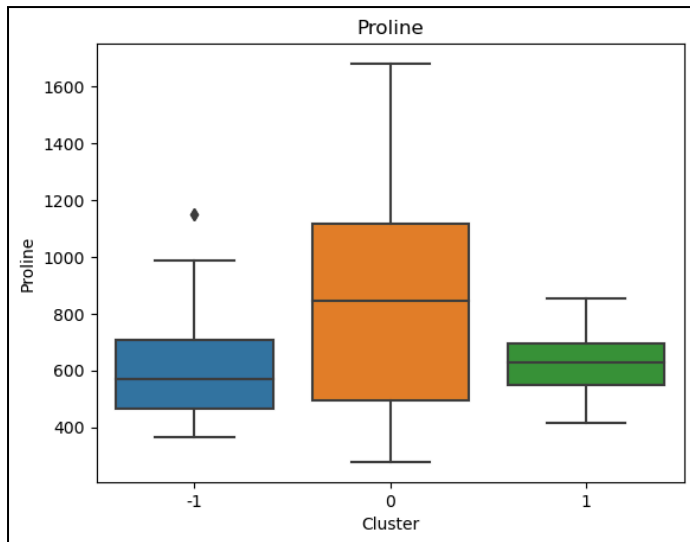












Interpretation:

- Cluster 0:
This cluster has the highest Proline values, indicating wines with stronger or richer chemical composition.
- Cluster 1:
This cluster has lower Proline but higher Alcalinity compared to Cluster 0, suggesting a different chemical profile.
- Cluster -1 (Noise):
This cluster represents outliers that do not fit into any main group.
This demonstrates DBSCAN's strength in identifying non-homogeneous data points.

Step 9: Conclusion

Interpretation:

The analysis demonstrates that DBSCAN is the most suitable clustering method for the wine dataset. Its ability to identify both structured clusters and outliers provides a more realistic representation of the data. The results also highlight that variables such as Proline play a significant role in differentiating clusters, indicating their importance in defining wine characteristics. Overall, clustering reveals meaningful patterns within the dataset and provides valuable insights into the underlying structure of the data.